ABSTRACT

The Smart Spell Checker is a dictionary-based query correction system designed to identify and correct spelling errors in user-input queries. In many applications, such as search engines, text editors, messaging platforms, and online forms, typing errors can prevent users from obtaining the intended results. Traditional exact-match systems fail to recognize words containing spelling mistakes, even when the entered words are very similar to the correct ones. This project addresses this problem by implementing an approximate string-matching technique that identifies the closest valid words from a predefined dictionary.

The system uses a dictionary data structure to store valid words and applies the **Levenshtein Edit Distance algorithm** to measure the difference between a user-entered query and dictionary words. The algorithm uses dynamic programming to calculate the minimum number of character insertions, deletions, and substitutions required to transform one word into another. A predefined maximum edit distance threshold is used to filter out unrelated words and retain relevant correction candidates. The resulting suggestions are ranked in ascending order of edit distance, allowing the system to present the closest matches first.

The system also incorporates query preprocessing to improve consistency and evaluates its behaviour using sample misspelled words. Test cases such as “recieve,” “wrold,” and “definately” demonstrate its ability to identify intended words, including “receive,” “world,” and “definitely,” respectively. The project illustrates the practical application of fundamental data structures, string processing, and dynamic programming in solving a real-world problem. The proposed system can be further enhanced through word-frequency ranking, advanced dictionary indexing techniques, multilingual support, and integration with real-time search and autocomplete applications.

Keywords: Spell Checking, Dictionary Data Structure, Levenshtein Edit Distance, Dynamic Programming, Query Correction, String Matching.